

STEFF BECKERS

IIS Express 10 localhost SSL certificate reset_

26/03/2021 21:48 / #dev #tutorial #dotnet #visual-studio #iis-express #ssl / 3 min read / 4 Comments

Tutorial on how to reset your localhost SSL certificate for IIS Express 10.

Source code

Quick test .NET Web API project to test the problem, fix the issue and check if it's working again.

<https://github.com/steffbeckers/iis-express-ssl-reset-test>

Problem

While running a test .NET Web API using IIS Express, the <https://localhost:...> site can't be reached or some other error like "invalid SSL certificate".



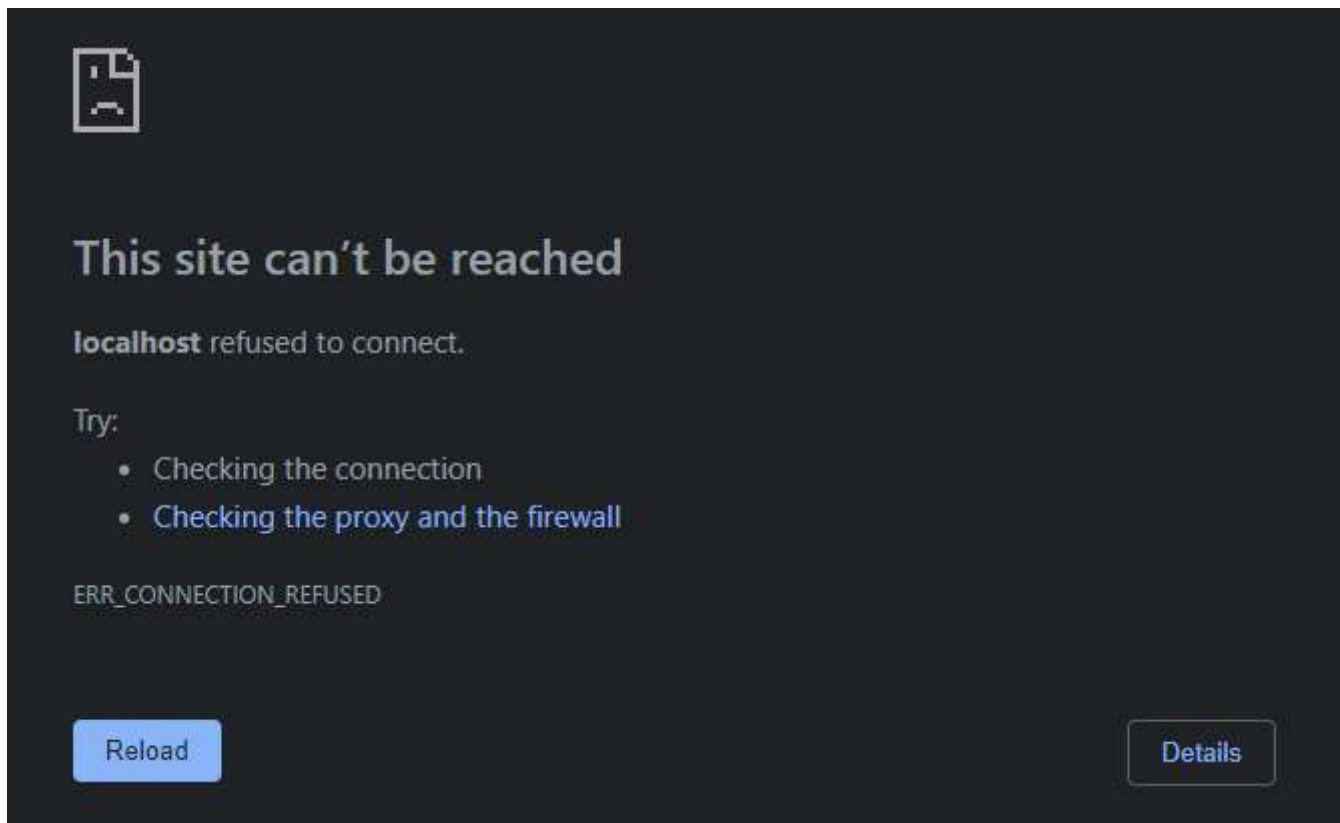
The screenshot shows the Visual Studio IDE with the following components:

- Menu Bar:** File, Edit, View, Git, Project, Build, Debug, Test, Analyze, Tools, Extensions, Window, Help.
- Search Bar:** Search (Ctrl+Q)
- Server Explorer:** Shows the project structure for 'Solution Test' (1 of 1 project).
 - Test
 - Connected Services
 - Dependencies
 - Properties
 - launchSettings.json (selected)
 - Controllers
 - WeatherForecastController.cs
 - Models
 - WeatherForecast.cs
 - appsettings.json
 - Program.cs
 - Startup.cs

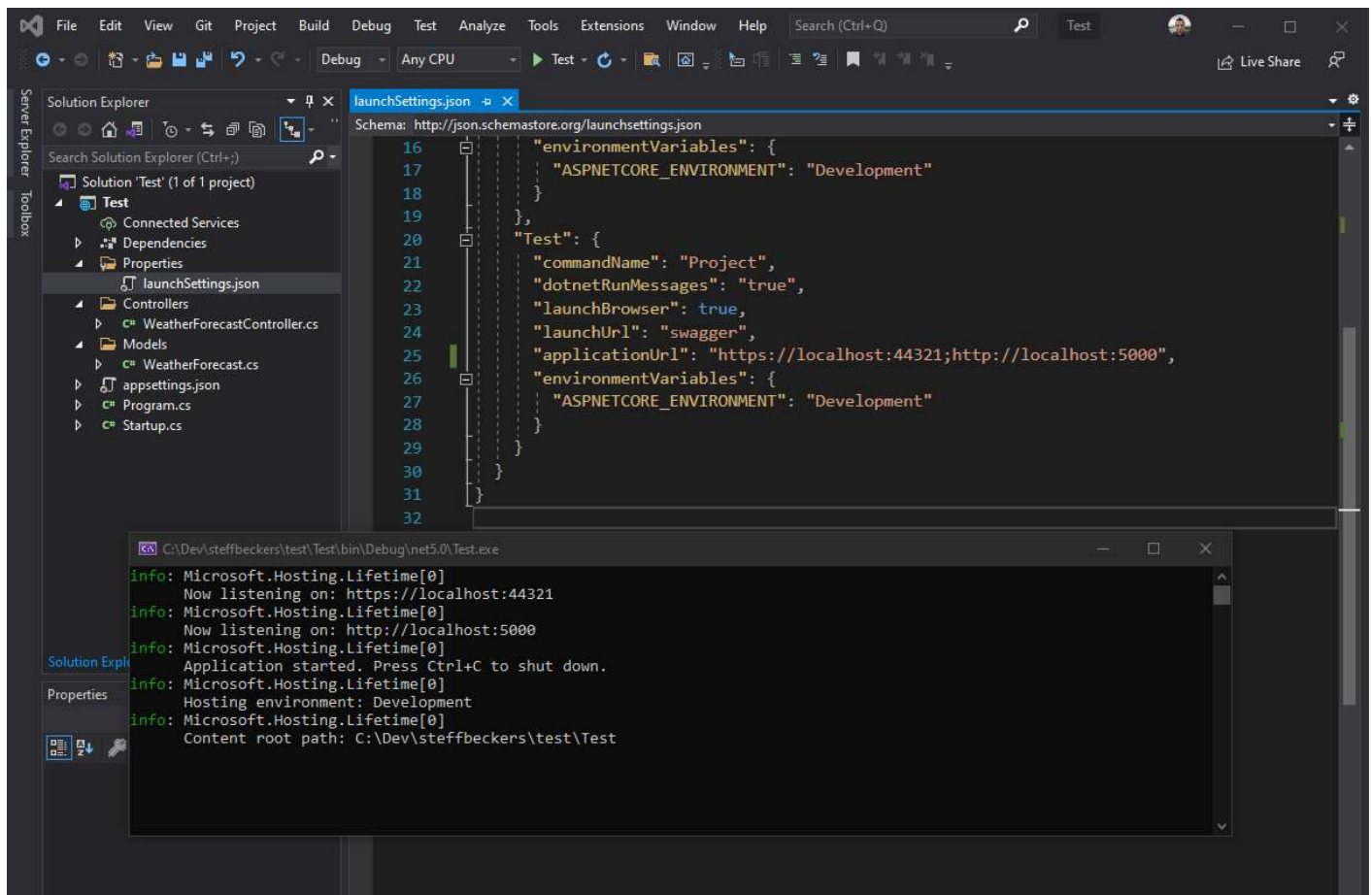
- Properties Window:** Empty.
- Code Editor:** Displays the content of `launchSettings.json`. The schema is `http://json.schemastore.org/launchsettings.json`.

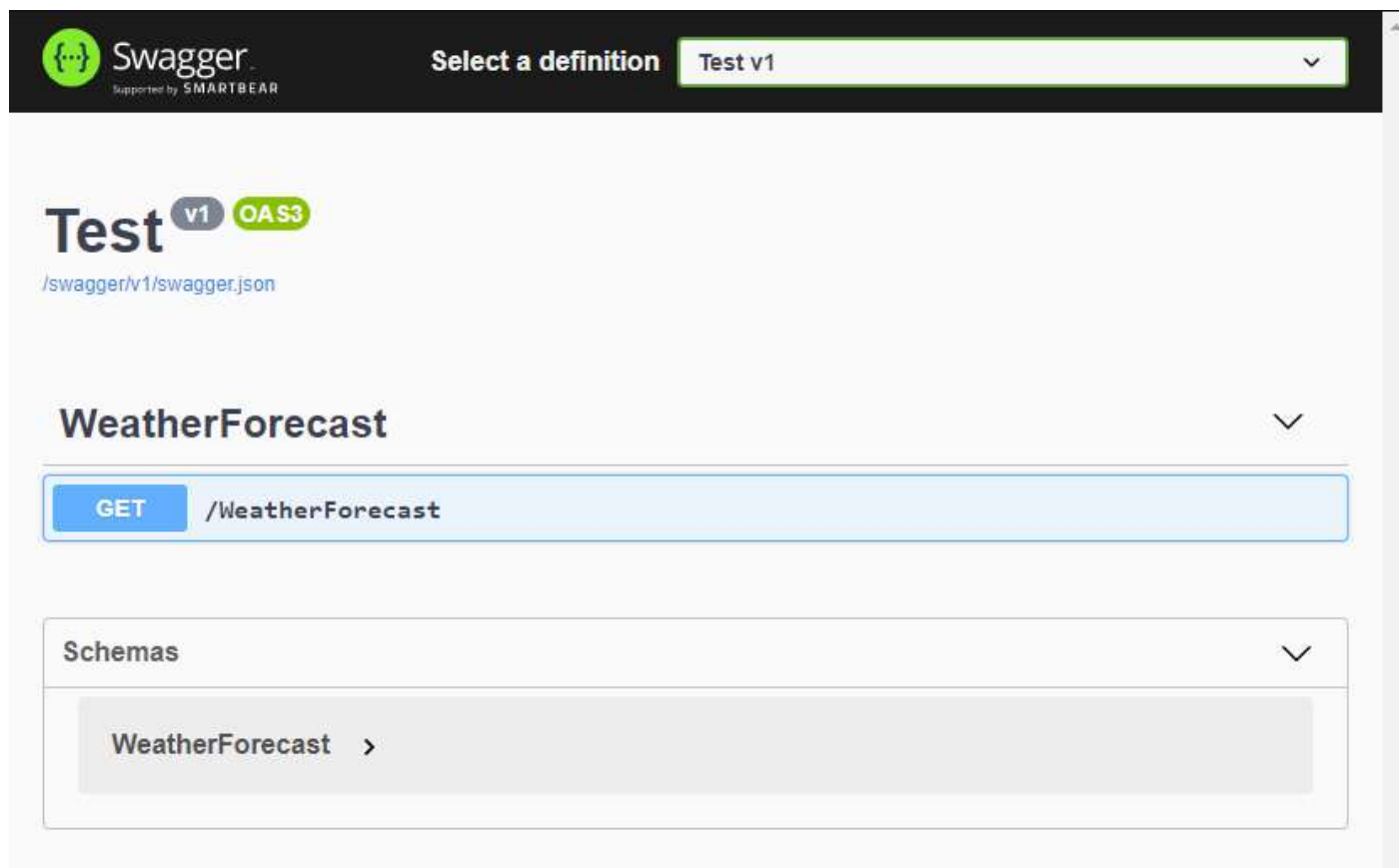
```
1 {
2   "$schema": "http://json.schemastore.org/launchsettings.json",
3   "iisSettings": {
4     "windowsAuthentication": false,
5     "anonymousAuthentication": true,
6     "iisExpress": {
7       "applicationUrl": "https://localhost:44321",
8       "sslPort": 44321
9     }
10  },
11  "profiles": {
12    "IIS Express": {
13      "commandName": "IISExpress",
14      "launchBrowser": true,
15      "launchUrl": "swagger",
16      "environmentVariables": {
17        "ASPNETCORE_ENVIRONMENT": "Development"
18      }
19    },
20    "Test": {
21      "commandName": "Project",
22      "dotnetRunMessages": "true",
23      "launchBrowser": true,
24      "launchUrl": "swagger",
25      "applicationUrl": "https://localhost:44321;http://localhost:5000",
26      "environmentVariables": {
27        "ASPNETCORE_ENVIRONMENT": "Development"
28      }
29    }
30  }
31 }
32 }
```
- Output Window:** Shows output from 'Test - ASP.NET Core Web Server'.

```
Starting IIS Express ...
Successfully registered URL "http://localhost:15190/" for site "Test" application "/"
Successfully registered URL "https://localhost:44321/" for site "Test" application "/"
Registration completed for site "Test"
IIS Express is running.
```
- Status Bar:** Ready. Add to Source Control.



But if we run the API with Kestrel, the site just works fine.



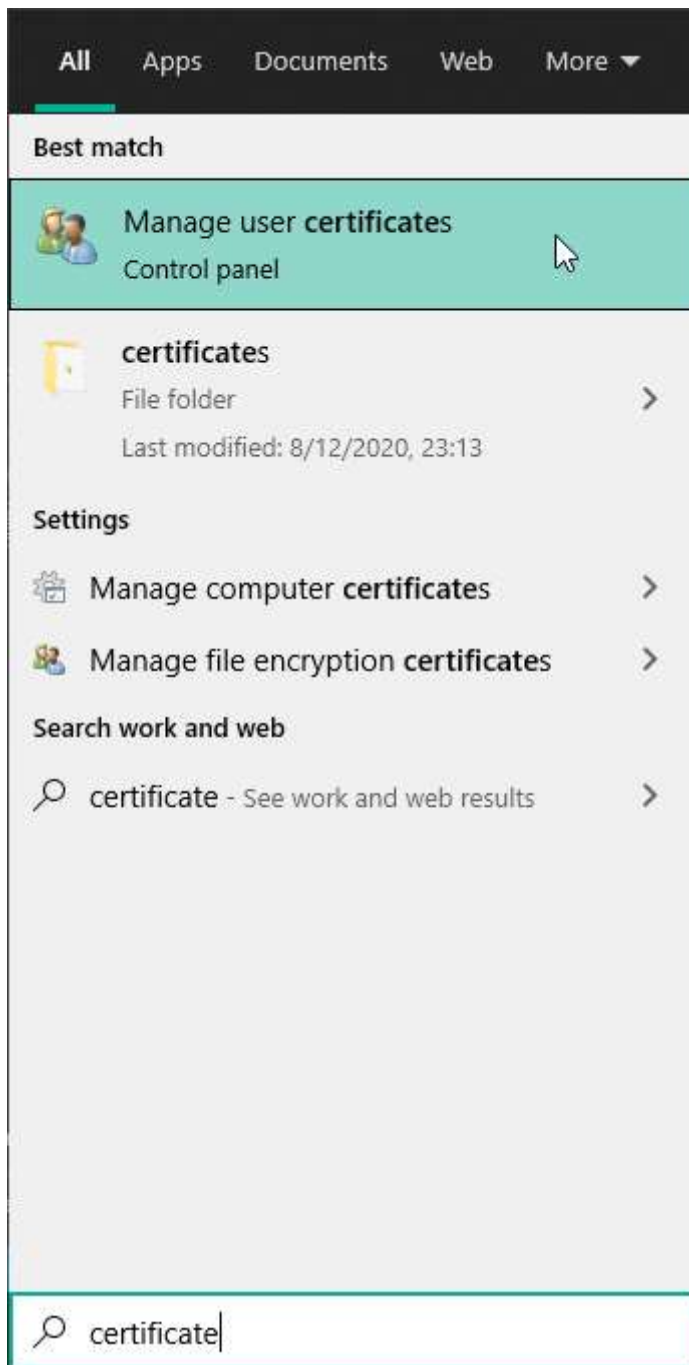


Or maybe this isn't even working for you.

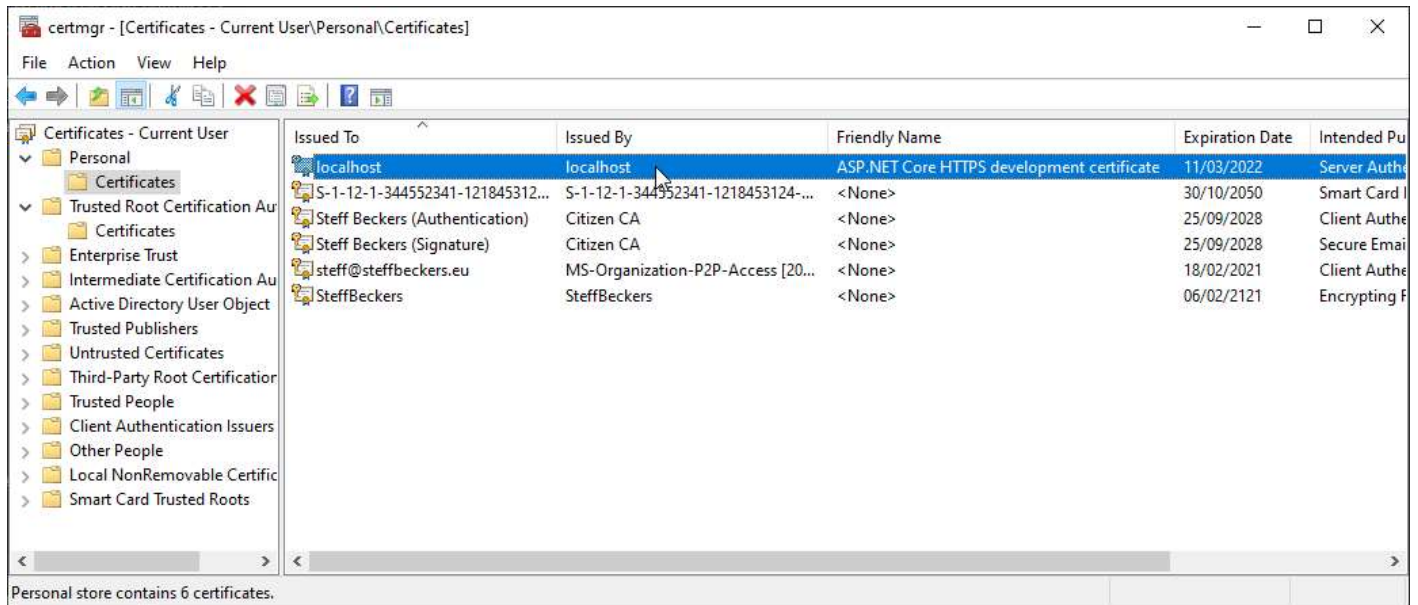
Solution

Step 1: Cleanup all localhost certificates

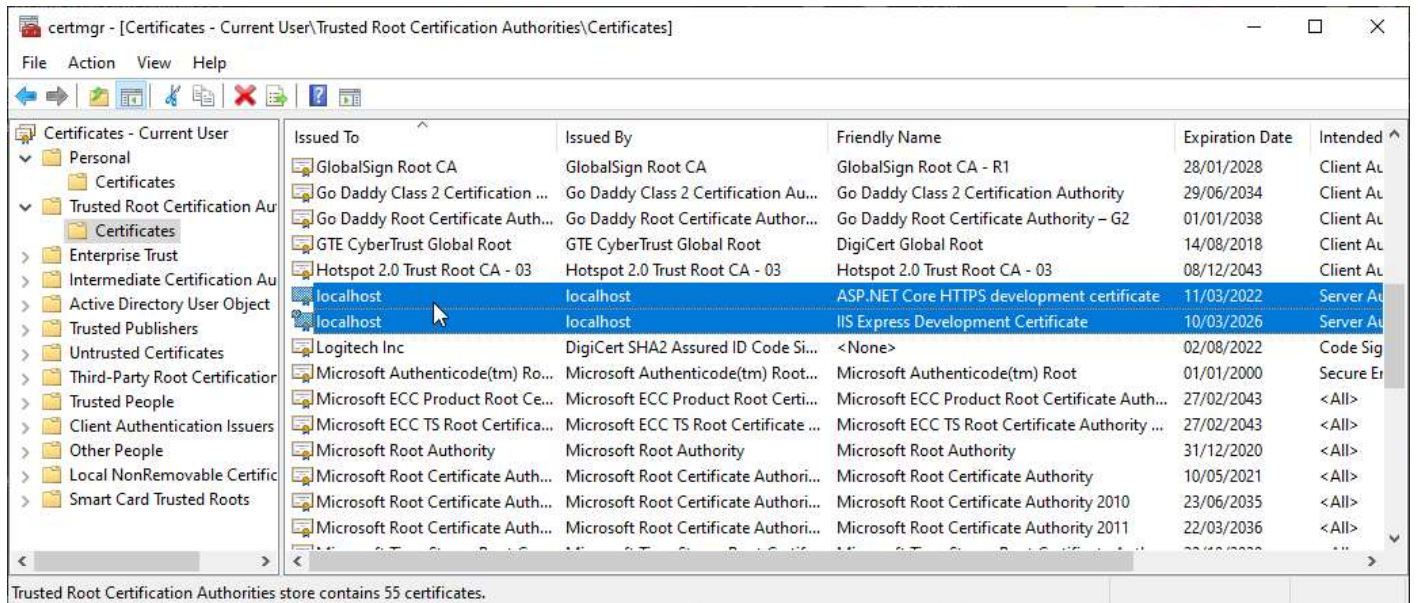
Open the Windows Certificate Manager by searching for "certificate". Choose "Manage user certificates".



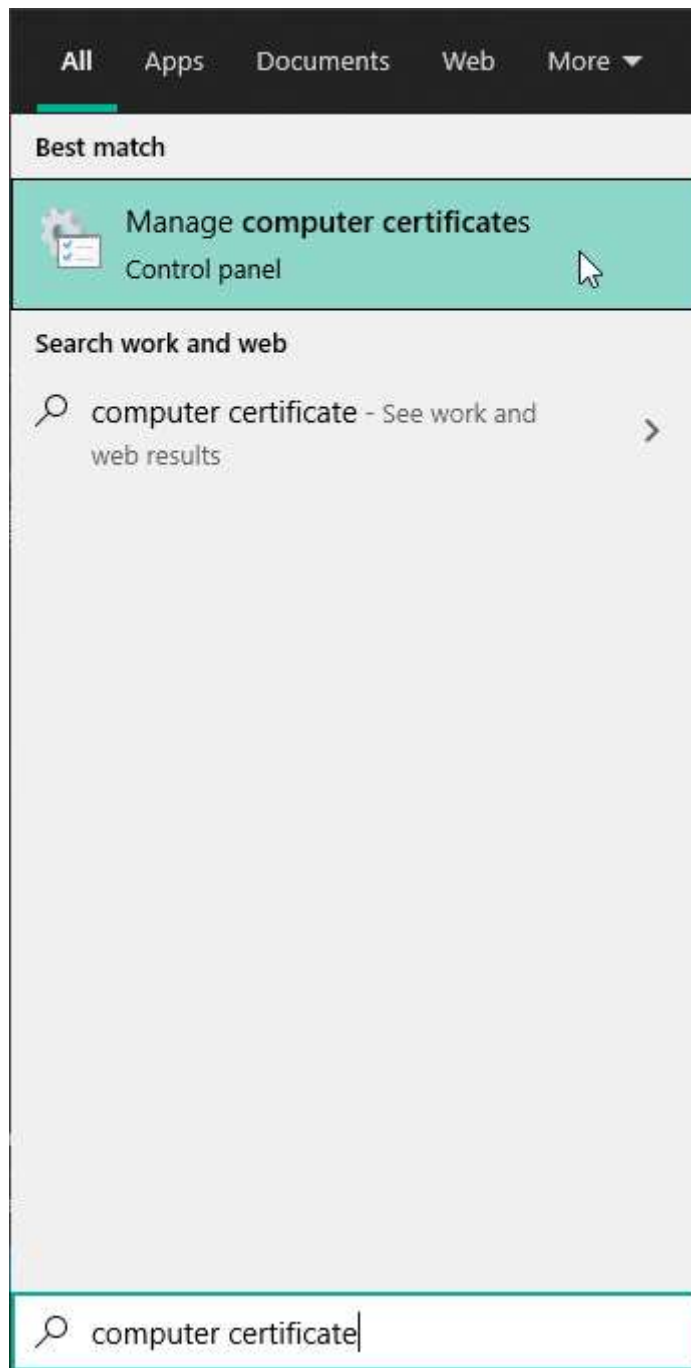
Select all localhost certificates in the "Personal/Certificates" folder and delete them.

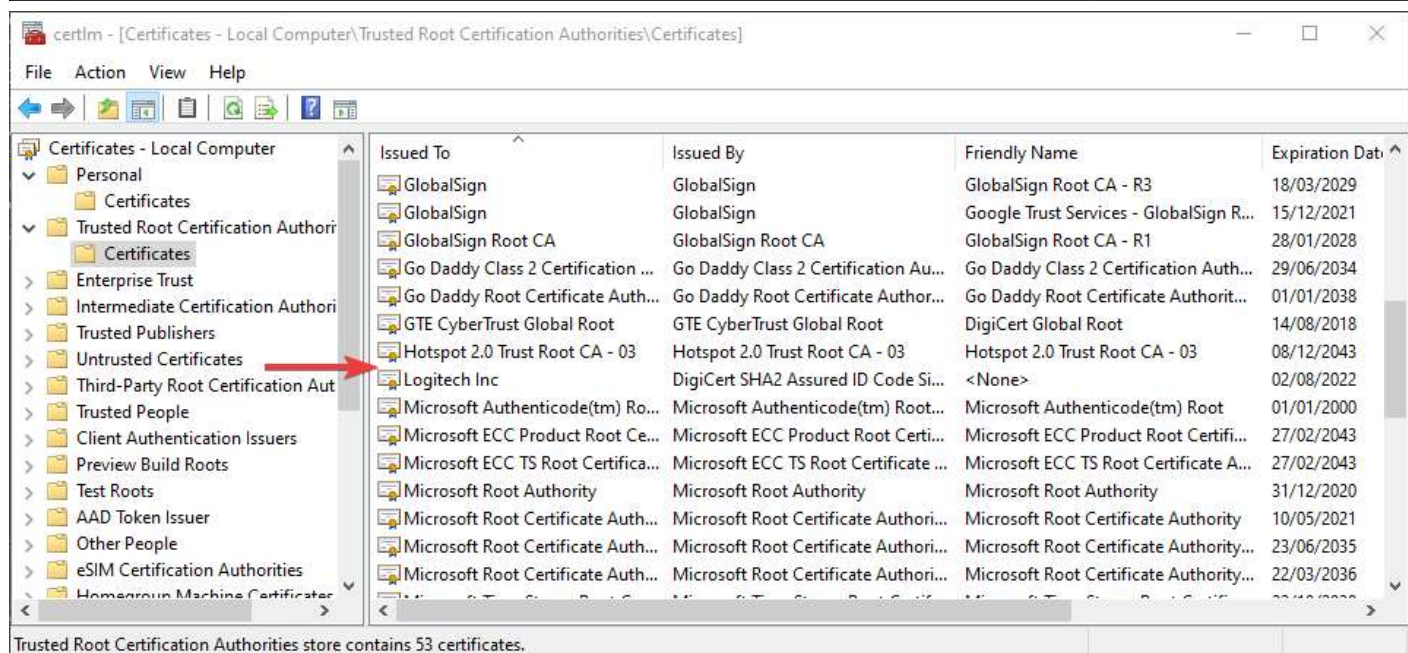
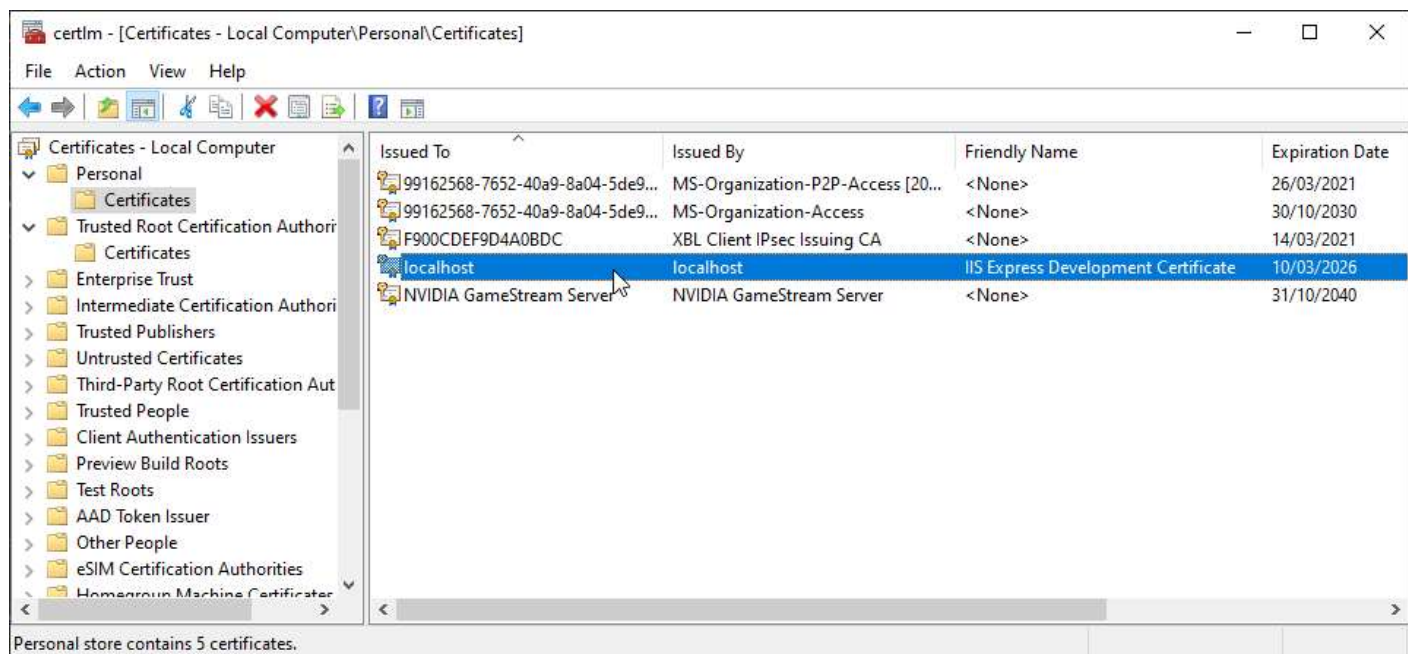


Select all localhost certificates in the "Trusted Root Certification Authorities/Certificates" folder and delete them.



Now do the same thing but for your local computer certificates.





Step 2: Use the IIS Express admin CLI

Open a new powershell or cmd prompt as Administrator (elevated permissions).

Change directory to your IIS Express installation folder.

```
cd "C:\Program Files (x86)\IIS Express"
```

Now execute the following command to re-run the setup for the SSL certificates.

Make sure to **update the port number** to match yours

```
./IisExpressAdminCmd.exe setupsslUrl -url:https://localhost:44321/ -UseSelfSig
```


Should give the following output:

```
C:\Program Files (x86)\IIS Express>
~ cd "C:\Program Files (x86)\IIS Express"
IIS Express .\IisExpressAdminCmd.exe setupsslUrl -url:https://localhost:44321/ -UseSelfSigned
Command 'setupsslUrl' completed.
IIS Express
```

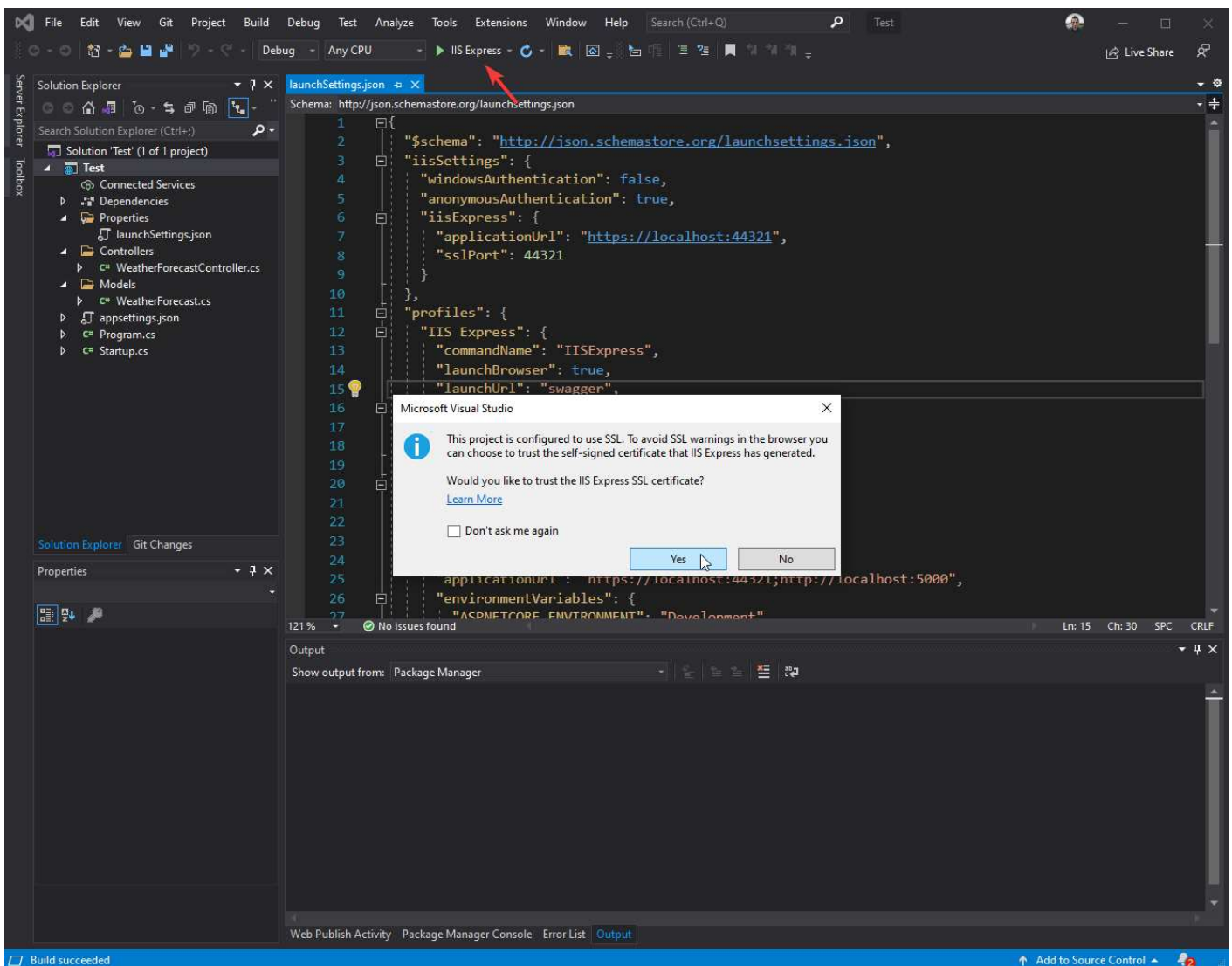
Step 3: Restart Visual Studio

Make sure to restart Visual Studio, this will also stop IIS Express.

Step 4: Re-run your application with IIS Express

Select IIS Express in your debug/run configuration and start the application.

Now you should get a pop-up to trust the new self-signed SSL certificate for your localhost app, we generated/setup in step 2. Choose "Yes" to trust the certificate.

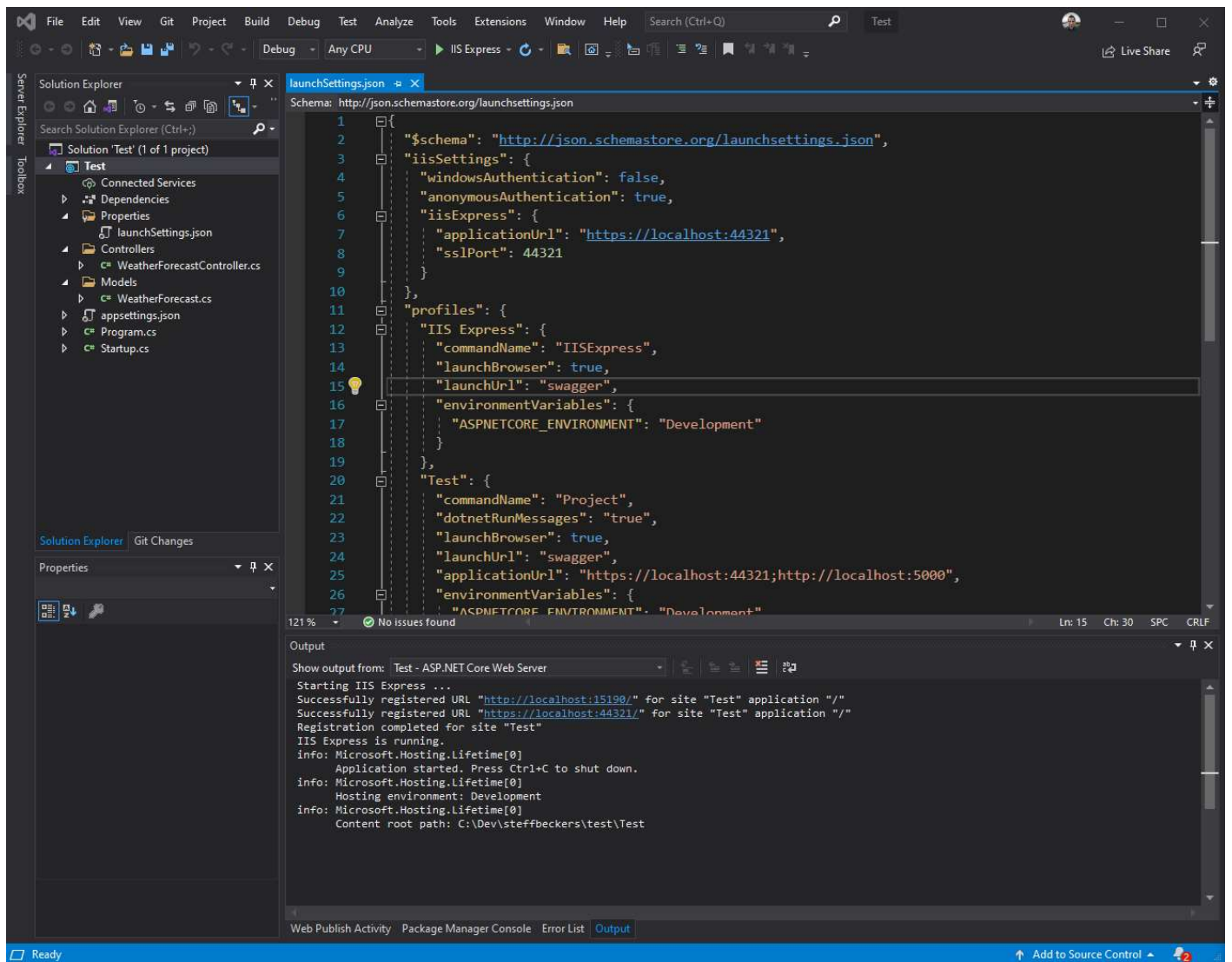


This pop-up should follow. This is a security warning you get before installing new certificates on your local machine. Choose "Yes" to install the certificate.



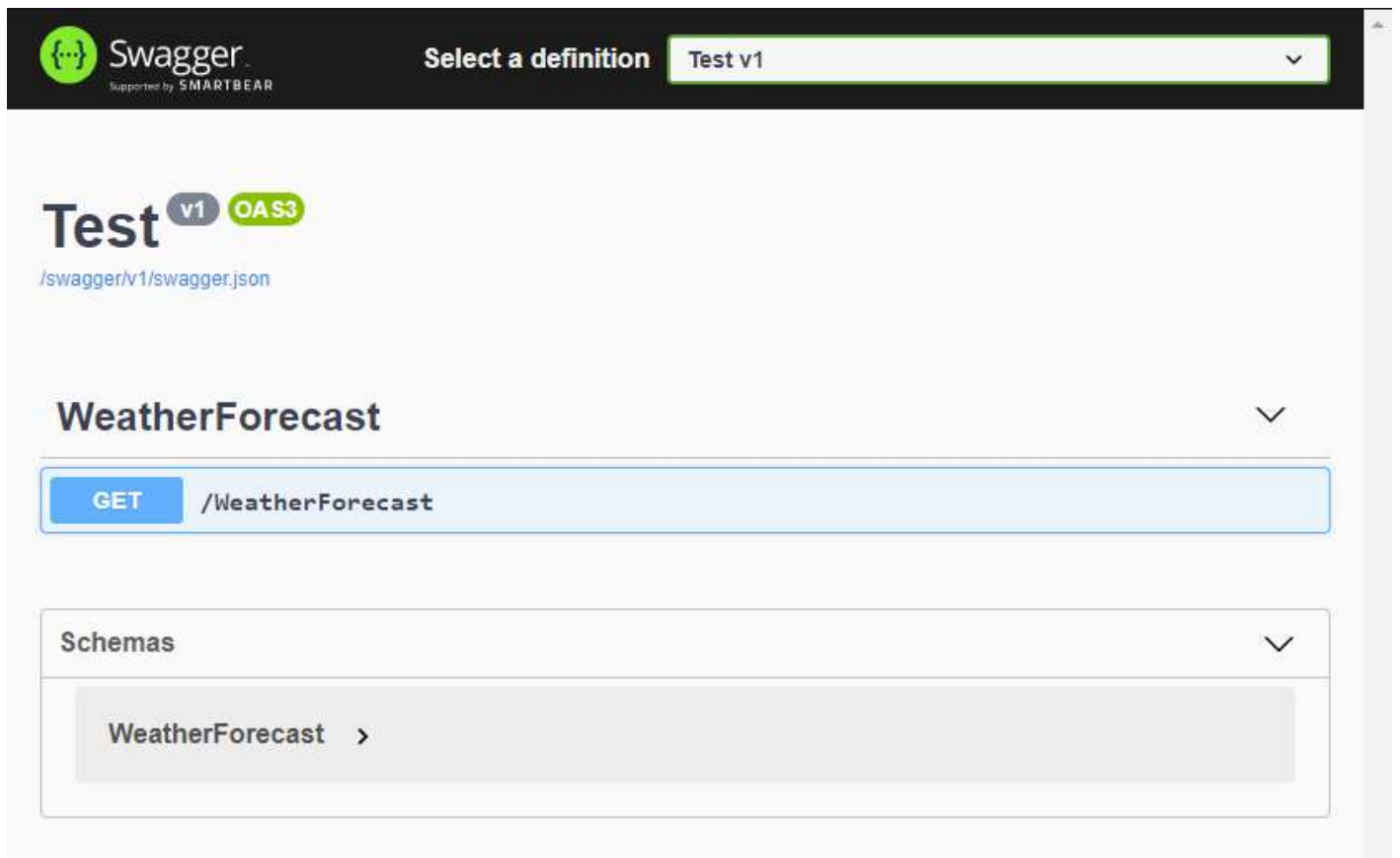
After installing the certificate the app should start correctly and the SSL problem should be fixed.

Now we get the "Application started. ..." output, in the Output window within Visual Studio.



Navigate/browse to the web page to test if the SSL certificate is fixed.

<https://localhost:44321/swagger>



Edit 31/03/2021

Also check out this GitHub Gist:

<https://gist.github.com/camieleggermont/5b2971a96e80a658863106b21c479988>

This powershell script generates a new certificate, removes the old certificate assignments from the IISExpress ssl ports and adds the newly generated one. The certificate is also copied over to the Trusted Root Certificate Authorities.

```
$cert = New-SelfSignedCertificate -DnsName "localhost", "localhost" -CertStore
$thumb = $cert.GetCertHashString()
```

```
For ($i=44300; $i -le 44399; $i++) {
    netsh http delete sslcert ipport=0.0.0.0:$i
}
```

```
For ($i=44300; $i -le 44399; $i++) {
    netsh http add sslcert ipport=0.0.0.0:$i certhash=$thumb appid='{214124cd-
}
```

```
$StoreScope = 'LocalMachine'
$StoreName = 'root'
```

```
$Store = New-Object -TypeName System.Security.Cryptography.X509Certificates.X  
$Store.Open([System.Security.Cryptography.X509Certificates.OpenFlags]::ReadWri  
$Store.Add($cert)  
  
$Store.Close()
```

This can help to reset all SSL certificates within the IIS Express port range 44300-44399. Make sure to open a powershell prompt as Administrator (elevated permissions).

Leave a comment or question



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